(Zane) Ziyang Ling

zaneling@mail.ustc.edu.cn | zaneling.github.io

Research Interests

Biomedical engineering master's student working on biomedical NLP, large language model reasoning, and medical agents. My research focuses on developing reliable, interpretable AI methods for biomedical text understanding and multi-agent system. I am interested in **clinical AI/NLP, clinical decision support and medical agents**.

Education

M.S. in Biomedical Engineering, University of Science and Technology of China (USTC) — Sep 2024 – Present

B.S. in Computer Science, Central China Normal University (CCNU) — Sep 2015 – Jun 2019

Research Projects

Biomedical Knowledge Graph Construction and Reasoning — Hefei, China

- Developed THOR[1] to mitigate error accumulation and attention drift in general multi-hop reasoning, and Beyond Reachability[2] to address semantically invalid paths between reachable nodes in biomedical knowledge graphs.
- Developed DualKL[3] to apply trust-region constraints to LoRA updates during training on biomedical corpora.
- Co-developed retrieval-augmented entity linking with BioNEL-RAG[4] and relation extraction based on LLM distillation with RoTDS-BioRE[5], and applied these methods to help construct the laboratory's retinal organoid knowledge graph.

Clinical Intuition Benchmarking and Multi-Agent Prescription Review — Beijing, China

- Investigating System 1 clinical intuition during a research internship at ByteDance by developing a benchmark to examine alignment between physicians' and language models' intuitive judgments. Starting from the dynamic clinical process, analyzed how models acquire evidence and update diagnoses as cases unfold to understand gaps in their clinical reasoning.[6]
- Developed PRISMATIC[7] with collaborating specialist agents that review prescriptions and identify potential medication conflicts across coexisting conditions, supporting physicians in making safer treatment decisions.

BioMARS: Multi-Agent Experiment Planning [Website] — Hefei, China

BioMARS is a multi-agent robotic platform that integrates scientific planning, robotic execution, and experimental feedback to automate biological experiments.[8]

- Led the design and development of the multi-agent system for BioMARS.
- Co-developed FGSS-BO[9] to address limitations of Bayesian optimization in selecting conditions for successive rounds of biological experiments. Incorporated biomedical knowledge as priors and used experimental feedback to guide hypothesis management and subsequent experiment selection.
- Developed the protocol-generation workflow and co-developed BioPRM[10] for step-level consistency verification, using verification feedback to evaluate and refine generated protocols before experimental execution.

Retinal Organoid Research and Data Platforms — Suzhou, China

- Built Organoid Chat and the Biomedical Data Interaction Platform as one integrated system (2025–2026), combining retrieval-augmented Q&A, search, and database linking to help researchers across groups explore organoid experiments, protocols, and data.
- Developed a Segment Anything Model (SAM)-based application for mouse retinal organoid segmentation (2025–2026), providing segmentation outputs for downstream analysis of organoid morphology.
- Responsible for the technical development of the multi-agent system in RAMP[11], the Researcher-Agent Modeling Pipeline for retinal organoid morphogenesis.

Selected Publications

[1] **Ziyang Ling**, Ronald X. Xu, Mingzhai Sun. THOR: A Theta-Gamma Hierarchical Oscillatory Reasoning Framework for Multi-hop QA. Accepted by *ACL 2026 Main*. 2026. [Paper] [Code]

[2] **Ziyang Ling**, Ziyi Chen, Ronald X. Xu, Mingzhai Sun. Beyond Reachability: A Relation-Semantic Reasoning Framework for Biomedical Multi-hop QA. Under review at *WWW*. 2026. [Paper]

[3] **Ziyang Ling**, Ronald X. Xu, Mingzhai Sun. DualKL: Distributional Trust Region for Continual LoRA Adaptation. Under review at *ICLR*. 2026.

[4] Ziyi Chen, Hui Liu, **Ziyang Ling**, Ronald X. Xu, Mingzhai Sun. Training-Free BioNEL-RAG: Advancing Biomedical Entity Linking through Retrieval-Augmented Generation and Large Language Models. Accepted by *ICHCI 2025*. 2025. [Paper]

[5] Hui Liu, **Ziyang Ling**, Ziyi Chen, Ronald X. Xu, Mingzhai Sun. RoTDS-BioRE: RoT-distillation of LLMs for semi-supervised biomedical relation extraction. Submitted to *BMC Bioinformatics*. 2026.

[6] **Ziyang Ling**. From Hindsight to Foresight: Dissecting Clinical Evidence Acquisition in Large Language Models. arXiv preprint; under review at *ICLR*. 2026. [Paper]

[7] **Ziyang Ling**, Shuyu Lu. PRISMATIC: Prescription Risk Inspection System for Multi-Agent Treatment Interaction. arXiv preprint. 2025. [Paper] [Code]

[8] Yibo Qiu, Zan Huang, Yilin Qiao, **Ziyang Ling**, et al. BioMARS: A Multi-Agent Robotic System for Autonomous Biological Experiments. Submitted to *Nature*. 2026.

[9] Hao Chen, **Ziyang Ling**, Ronald X. Xu, Mingzhai Sun. FGSS-BO: Feedback-Guided Hypothesis Management for LLM-Based Bayesian Optimization. arXiv preprint; under review at *COLING*. 2026. [Paper]

[10] Bowen Shao, **Ziyang Ling**, Ronald Xu, Mingzhai Sun. BioPRM: Step-Level Consistency Verification of Biological Protocols. arXiv preprint; under review at *COLING*. 2026. [Paper]

[11] Junao Song, Yi Wang, Yilin Qiao, Haoran Zhang, Mohan Yang, Fenghao Gu, Wenteng Gao, **Ziyang Ling**. RAMP: Researcher-Agent Modeling Pipeline for Retinal Organoid Morphogenesis. Submitted to *IEEE Journal of Biomedical and Health Informatics*. 2026.

Work Experience

ByteDance | Research Intern | Beijing, China — Aug 2026 – Present

Clinical intuition benchmarks and medical safety review agents.

Midea Group | Research Intern | Hefei, China — 2021 – 2024

Developed computer vision for refrigerator IoT and explored AI agents connecting food information with human health.

Yangling Technology Co., Ltd. | Software Engineer | Hefei, China — 2019 – 2020

Honors and Leadership

First-Class Academic Scholarship, USTC — 2026

Gold Award, Hubei Provincial Innovation and Entrepreneurship Competition (University-Level) — 2018

University President of the Student Union, CCNU — 2018